

HIBA MADIHA LASHARI

Computer Vision / AI Engineer

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PROFILE

Innovative **Computer Vision and Artificial Intelligence Engineer** with 3+ years of experience in designing, developing, testing, and deploying deep learning models for real-time applications. Strong expertise in **object detection, image classification, tracking, and edge AI optimization**. Experienced in translating research into **production-ready AI systems** for agriculture, surveillance, and industrial inspection. Proficient in Python-based AI development, modern deep learning frameworks, and deployment on embedded and cloud platforms.

EXPERIENCE

Artificial Intelligence Technology Centre (AITeC), National Centre for Physics
Assistant Manager AI (Full-time) **04.2023 – Present**
Federal Government of Pakistan

- Designed and deployed computer vision systems for real-time detection and classification tasks.
- Worked extensively on YOLO-based object detection, image classification, and tracking pipelines.
- Performed model optimization (pruning, quantization) for deployment on edge and embedded devices.
- Developed AI solutions for IC defect detection, crop disease classification, and facial emotion recognition.
- Built end-to-end ML workflows including data acquisition, preprocessing, training, evaluation, and deployment.
- Supervised interns and contributed to national AI seminars and hands-on training programs.

Capital University of Science & Technology (CUST), Islamabad
Research Assistant / Lab Engineer (Full-time) **10.2020 – 06.2022**

- Implemented scale-invariant object tracking algorithms for thermal infrared video sequences.
- Addressed challenges such as occlusion, scale variation, and low-contrast imagery.
- Assisted in laboratory teaching for Introduction to Computing, Programming, Digital Logic Design, and Electronics.
- Mentored undergraduate students in computer vision and AI-based final year projects.

EDUCATION

Capital University of Science & Technology (CUST), Islamabad
Master of Science – Electrical Engineering **10.2020 – 06.2022**

- CGPA: **3.80 (1st Rank)**
- Thesis: *Scale-Invariant Tracking and Occlusion Detection in Thermal Infrared Sequences*
- Dean's Honor Award

Capital University of Science & Technology (CUST), Islamabad
Bachelor of Science – Electrical Engineering **08.2016 – 09.2020**

- Final Year Project: *Gesture-Controlled Gun Manoeuvring Using Vision and Servo Systems*
- Dean's Honor Award

RESEARCH PROJECTS

Object Detection on Raspberry Pi for IoT Surveillance

- Deployed YOLO-based models on Raspberry Pi for real-time video surveillance.
- Optimized inference using pruning, quantization, and resolution tuning.

- Integrated RTSP/IP camera feeds and streamed results to a monitoring dashboard.
- Evaluated multiple YOLO versions for speed-accuracy trade-offs on edge devices.

IC Defect Detection Using Deep Learning (YOLOv8)

- Built an automated inspection system for microchip surface defect detection.
- Achieved high precision in detecting minor defects for industrial quality assurance.

Wheat Disease Classification (ViT vs CNN)

- Conducted comparative analysis between Vision Transformers (ViT-B32) and CNNs.
- Achieved 99.2% accuracy using ViT, outperforming VGG-19.

Scale-Invariant Object Tracking in Thermal Infrared Sequences

- Developed a robust tracking algorithm handling scale variation and occlusion.
- Integrated Discriminative Correlation Filters with Kalman filtering.
- Improved tracking performance by approximately 25% over baseline methods.

RESEARCH PUBLICATIONS

- Scale Invariant Tracking of Objects and Occlusion Detection in Challenging Thermal Infrared Sequences, **IEEE INMIC 2023**
- ViT vs CNN: A Comparative Study of Wheat Disease Classification for Custom Data, **IEEE FIT 2023**

SKILLS IN ARTIFICIAL INTELLIGENCE

Computer Vision, Object Detection & Classification, Vision Transformers, Image Processing, Deep Learning, Model Optimization (Pruning, Quantization), Edge AI Deployment, Data Acquisition & Pre-processing

OTHER TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB

Frameworks: PyTorch, TensorFlow, Keras

Libraries: OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib

APIs & Deployment: FastAPI, Streamlit, Gradio

DevOps / MLOps: Docker, Kubernetes, Git, GitLab

Tools: Weights & Biases, Roboflow, Label Studio

Hardware: Raspberry pi, NVIDIA Jetson, GPUs, CUDA (working knowledge)

CERTIFICATIONS & TRAINING

- Advanced AI Fellowship – AITeC, NCP (ML, DL, Transformers, RL, GPU & CUDA, Cloud AI)
- Docker and Kubernetes Training – AITeC, NCP
- Hadoop Training – AITeC, NCP

ACADEMIC & PROFESSIONAL INVOLVEMENT

- Instructor in Basic and Advanced AI training programs (ML, DL, Generative AI).
- Organized and hosted National AI Seminars (NAIS) at AITeC-NCP.
- Hosted ISPAD 2024 Conference at AITeC-NCP.
- Mentored AI interns on applied research and development projects.